AUO Display+

■ **Preliminary Specification**
□ **Final Specification**

Module	21.5 Inch Color TFT-LCD with Touch
Model Name	G215HAT02.1
TP FW version	TBD

Company		**Approved by**	**Date**
		Elsie Kuo	2026/1/2
Checked & Approved by	**Date**	**Prepared by**	
		Eugene Yu	2026/1/2
		General Display Business Unit / AUO Display Plus Corporation	

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Record of Revision

Version and Date	Page	Old description	New Description
0.1 2026/1/2	All		1'st Edition

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1. Operating Precautions

1) Since front polarizer is easily damaged, please be cautious and not to scratch it.
2) Be sure to turn off power supply when inserting or disconnecting from input connector.
3) Wipe off water drop immediately. Long contact with water may cause discoloration or spots.
4) When the panel surface is soiled, wipe it with absorbent cotton or soft cloth.
5) Since the panel is made of glass, it may be broken or cracked if dropped or bumped on hard surface.
6) To avoid ESD (Electro Static Discharde) damage, be sure to ground yourself before handling TFT-LCD Module.
7) Do not open nor modify the module assembly.
8) Do not press the reflector sheet at the back of the module to any direction.
9) In case if a module has to be put back into the packing container slot after it was taken out from the container, do not press the center of the LED light bar edge. Instead, press at the far ends of the LED light bar edge softly. Otherwise the TFT Module may be damaged.
10) At the insertion or removal of the Signal Interface Connector, be sure not to rotate nor tilt the Interface Connector of the TFT Module.
11) TFT-LCD Module is not allowed to be twisted & bent even force is added on module in a very short time. Please design your display product well to avoid external force applying to module by end-user directly.
12) Small amount of materials having no flammability grade is used in the LCD module. The LCD module should be supplied by power complied with requirements of Limited Power Source (IEC60950 or UL1950), or be applied exemption.
13) Severe temperature condition may result in different luminance, response time and lamp ignition voltage.
14) Continuous operating TFT-LCD display under low temperature environment may accelerate lamp exhaustion and reduce luminance dramatically.
15) The data on this specification sheet is applicable when LCD module is placed in landscape position.
16) Continuous displaying fixed pattern may induce image sticking. It's recommended to use screen saver or shuffle content periodically if fixed pattern is displayed on the screen.

2. General Description

This specification applies to the 21.5 inch-wide Color TFT-LCD Module with Touch G215HAT02.1. The display supports the Full HD-1920(H)x1080(V) screen format and 16.7M colors. All input signals are LVDS interface comaptible.

2.1 Display Characteristics (G215HAT02.1)

The following items are characteristics summary on the table under 25 ℃ condition:

Items	Unit	Specifications
Screen Diagonal	[mm]	(21.5")
Active Area	[mm]	476.064 (H) x 267.786 (V)
Pixels H x V		1920(x3) x 1080
Pixel Pitch	[um]	247.95 (per one triad) ×247.95
Pixel Arrangement		R.G.B. Vertical Stripe
Display Mode		AHVA Mode, Normally Black
White Luminance (Center)	[cd/m²]	350 cd/m² (Typ.)
Contrast Ratio		1000 (Typ.)
Optical Response Time	[msec]	22 (Tr+Tf)
Nominal Input Voltage	[Volt]	5 V(Typ)
Power Consumption (VDD line + LED line)	[Watt]	LCD module: PDD (Typ.)= 2.5 @ White pattern, Fv=60Hz Backlight unit: PBLU=(10.03)W (Typ.)
Weight	[Grams]	2,530 g(Typ.)
Physical Size (TTL)	[mm]	491.62(H) x 291.22(V) x 15.35(D) Typ (excluded LED driver board)
Electrical Interface		Dual LVDS
Support Color		16.7M
Surface Treatment		AS coating
Temperature Range Operating Storage (Non-Operating)	 [ºC] [ºC]	 0 to +60 -20 to +60
RoHS Compliance		RoHS Compliance

2.2 General Touch Characteristics

Item		Unit	Specifications
Type			Projected Capacitive Touch Panel
Structure			Glass / Glass
Mounting type			Direct Bonding
Input Mode			Multi Finger
Cover lens	O.D.	[mm]	491.62(H) x 291.22(V)
	Thickness	[mm]	1.1
C/L Visual Area		[mm]	477.06 X 268.79
Sensor Glass	O.D.	[mm]	490.62(W)x284.27(H)
	Thickness	[mm]	0.7
TP Active Area		[mm]	479.66 x 271.39
Substrate Material			SDL CS Glass
Touch points			10
Transmittance (%)		%	≥ 87
Single Touch Point Life Time			1 Millions
Surface Hardness			7H
Interface			USB 2.0 full speed
Single / Multi-touch Accuracy		[mm]	Center:+/-1.5mm Edge: +/-2 mm
Linearity		[mm]	Center: +/-1.5mm Edge: +/-2mm
The smallest distance between 2 points (each point diameter is 7mm)		[mm]	15
Channel (X * Y)			80 * 45
Report Rate (points /sec)		[Hz]	>100
Power Consumption		[mW]	550(typ.)
Response Time		[ms]	<35
Operating System			Support Win8 / Win10 ,Linux & Android
TP Firmware Version			TBD

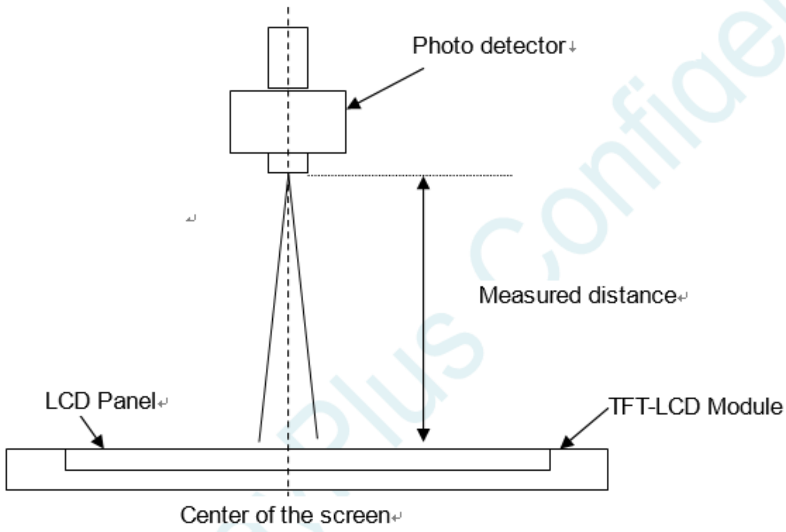
2.3 Optical Characteristics

The optical characteristics are measured under stable conditions at 25℃ (Room Temperature):

Item	Unit	Conditions		Min.	Typ.	Max.	Note
Viewing Angle	[degree]	Horizontal CR = 10	(Right)		89	-	1, 2
			(Left)		89	-	
	[degree]	Vertical CR = 10	(Upper)		89	-	
			(Lower)		89	-	
Contrast ratio		Normal Direction		600	1000	-	3
Response Time	[msec]	Gray To Gray			22		4
Color / Chromaticity Coordinates (CIE)		Red x		0.626	0.656	0.686	5
		Red y		0.298	0.328	0.358	
		Green x		0.263	0.293	0.323	
		Green y		0.584	0.614	0.644	
		Blue x		0.117	0.147	0.177	
		Blue y		0.037	0.067	0.097	
Color Coordinates (CIE) White		White x		0.283	0.313	0.343	
		White y		0.299	0.329	0.359	
Central Luminance	[cd/m^2]			280	350		6
Luminance Uniformity	[%]			80	85	-	7
Color Gamut	[%]	sRGB		95%	99%		

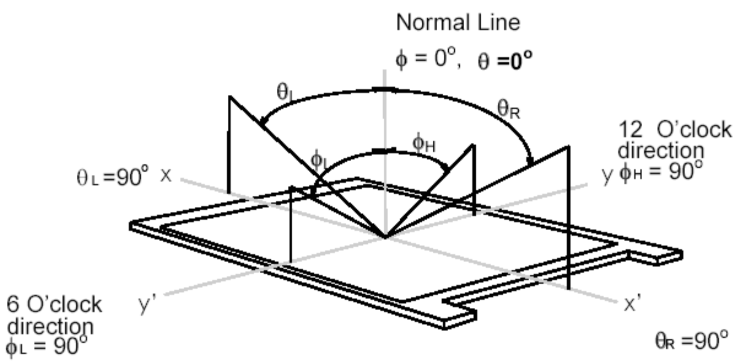
Note 1: Measurement method

Before measuring, the LCD module should be turn on 30 minutes at room temperature. In order to stabilize the luminance, the measurement should be executed after lighting Backlight for 30 minutes in a stable, windless and dark room.



Note 2: Definition of viewing angle measured by ELDIM (EZContrast 88)

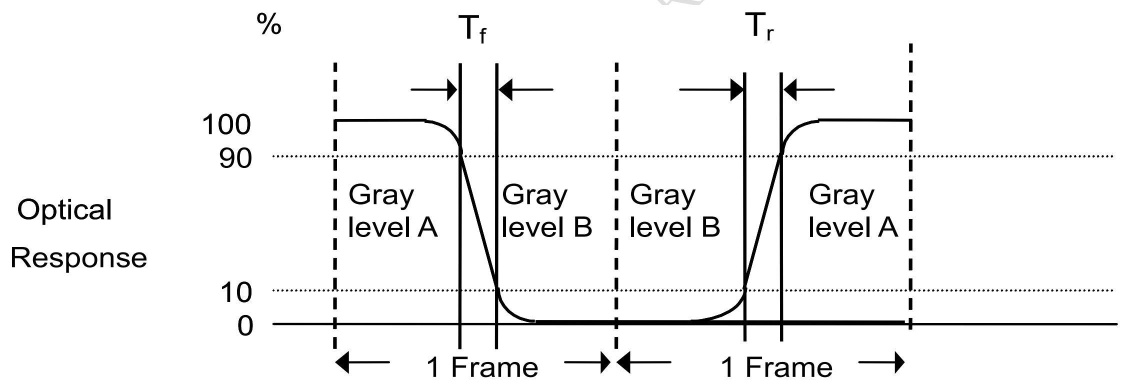
Viewing angle is the measurement of contrast ratio ≧ 10, at the screen center, over a 180° horizontal and 180° vertical range (off-normal viewing angles). The 180° viewing angle range is broken down as follows; 90° (θ) horizontal left and right and 90° (Φ) vertical, high (up) and low (down). The measurement direction is typically perpendicular to the display surface with the screen rotated about its center to develop the desired measurement viewing angle.



Note 3: Contrast ratio is measured by TOPCON SR-3

Note 4: Response time measurement

The output signals of photo detector are measured when the input signals are changed from "Gray level A" to "Gray level B" (falling time, TF), and from "Gray level B" to "Gray level A" (rising time, TR), respectively. The response time is interval between the 10% and 90% of optical response.



The gray to gray response time is defined as the following table.

Gray Level to Gray Level		***Target gray level***				
		L0	***L63***	***L127***	***L191***	***L255***
Start gray level	***L0***					
	L63					
	L127					
	L191					
	L255					

- T_{GTG_typ} is the total average time at rising time and falling time of gray to gray.

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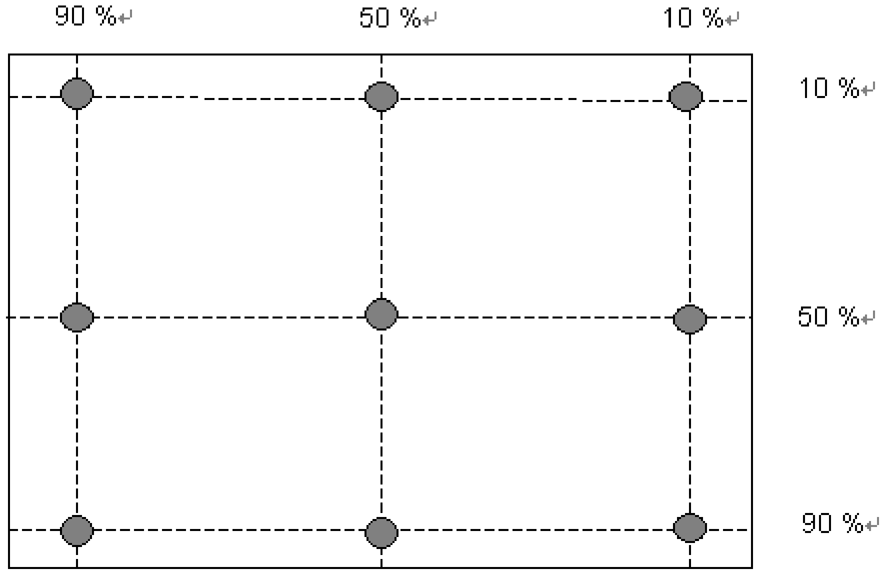
Note 5: Color chromaticity and coordinates (CIE) is measured by TOPCON SR-3

Note 6: Central luminance is measured by TOPCON SR-3

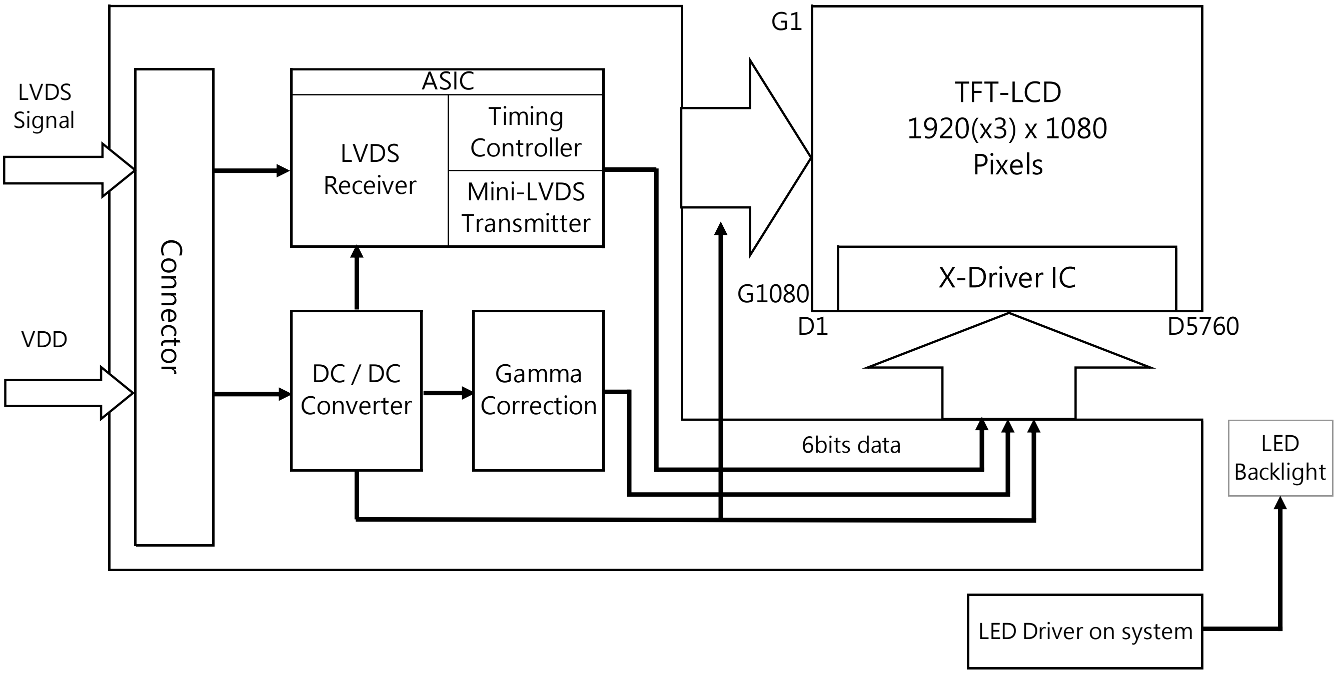
Note 7: Luminance uniformity of these 9 points is defined as below and measured by TOPCON SR-3



$$\text{Uniformity} = \frac{\text{Minimum Luminance in 9 points}(1-9)}{\text{Maximum Luminance in 9 Points}(1-9)}$$

3. Functional Block Diagram

The following diagram shows the functional block of this model.



4. Absolute Maximum Ratings

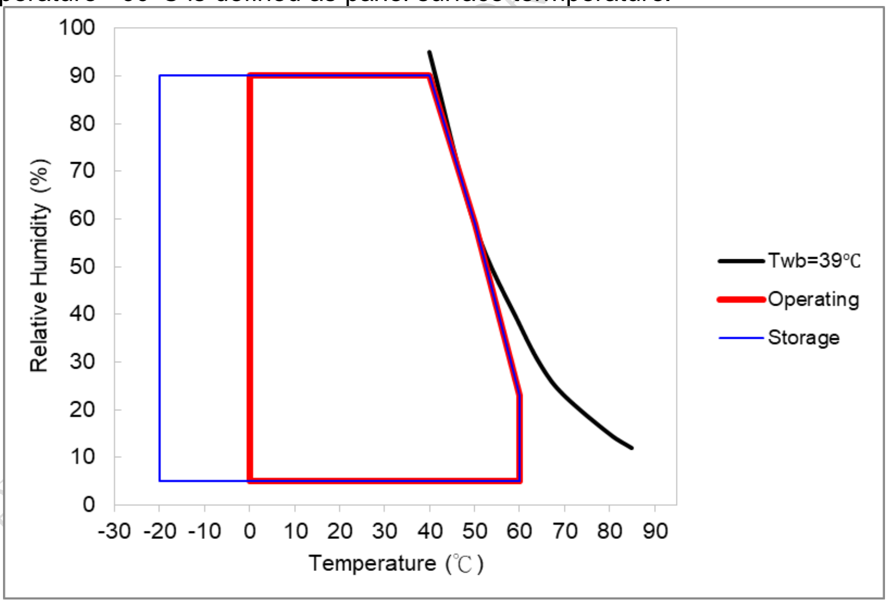
4.1 TFT LCD Module

Item	Symbol	Min	Max	Unit	Conditions
Logic/LCD Drive Voltage	VDD	0	5.5	[Volt]	Note 1,2

4.2 Absolute Ratings of Environment

Item	Symbol	Min.	Max.	Unit	Conditions
Operating Temperature	TOP	0	60	[°C]	Note 3 & 4
Operation Humidity	HOP	5	90	[%RH]	
Storage Temperature	TST	-20	60	[°C]	
Storage Humidity	HST	5	90	[%RH]	

Note 1: With in Ta (25 °C)
Note 2: Permanent damage to the device may occur if exceeding maximum values
Note 3: For quality perfermance, please refer to AUO IIS(Incoming Inspection Standard).
Note 4: Operation Temperature +60°C is defined as panel surface termperature.



5. Electrical Characteristics

5.1 TFT LCD Module

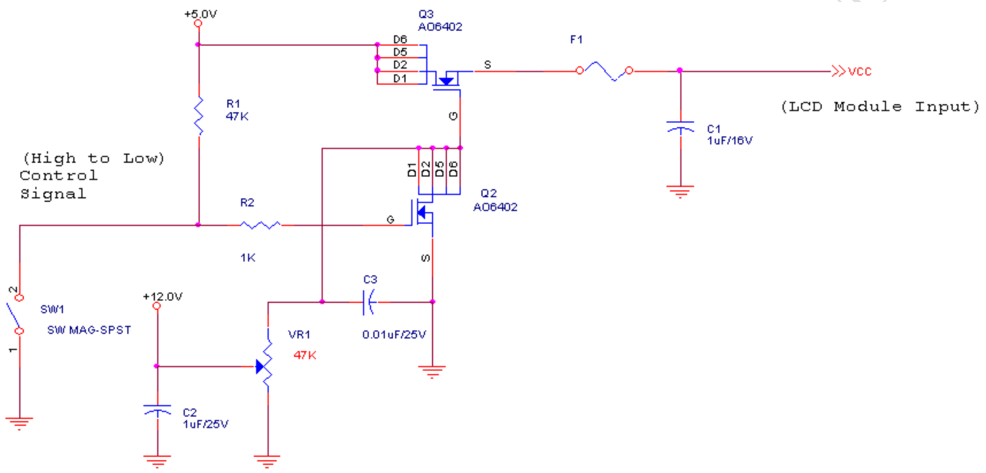
5.1.1 Power Specification

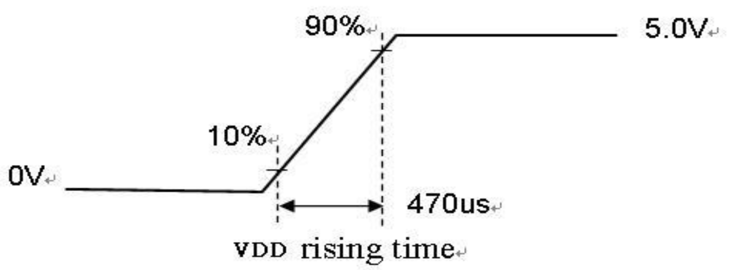
Input power specifications are as follows:

Symbol	Description	Min	Typ	Max	Unit	Remark
VDD	Input voltage	4.5	5.0	5.5	[Volt]	
IDD	Input Current (RMS)	-	0.5	0.6	[A]	VDD= 5.0V, White Pattern, Fv=60Hz
PDD	Consumption	-	2.5	3	[Watt]	VDD= 5.0V, White Pattern, Fv=60Hz
IRush	Inrush Current	-	-	3.0	[A]	Note 5-1
VDDrp	Allowable VDD Ripple Voltage	-	-	500	[mV]	VDD= 5.0V, White Pattern, Fv=60Hz

Note 1: Measurement conditions:

The duration of rising time of power input is 470us.



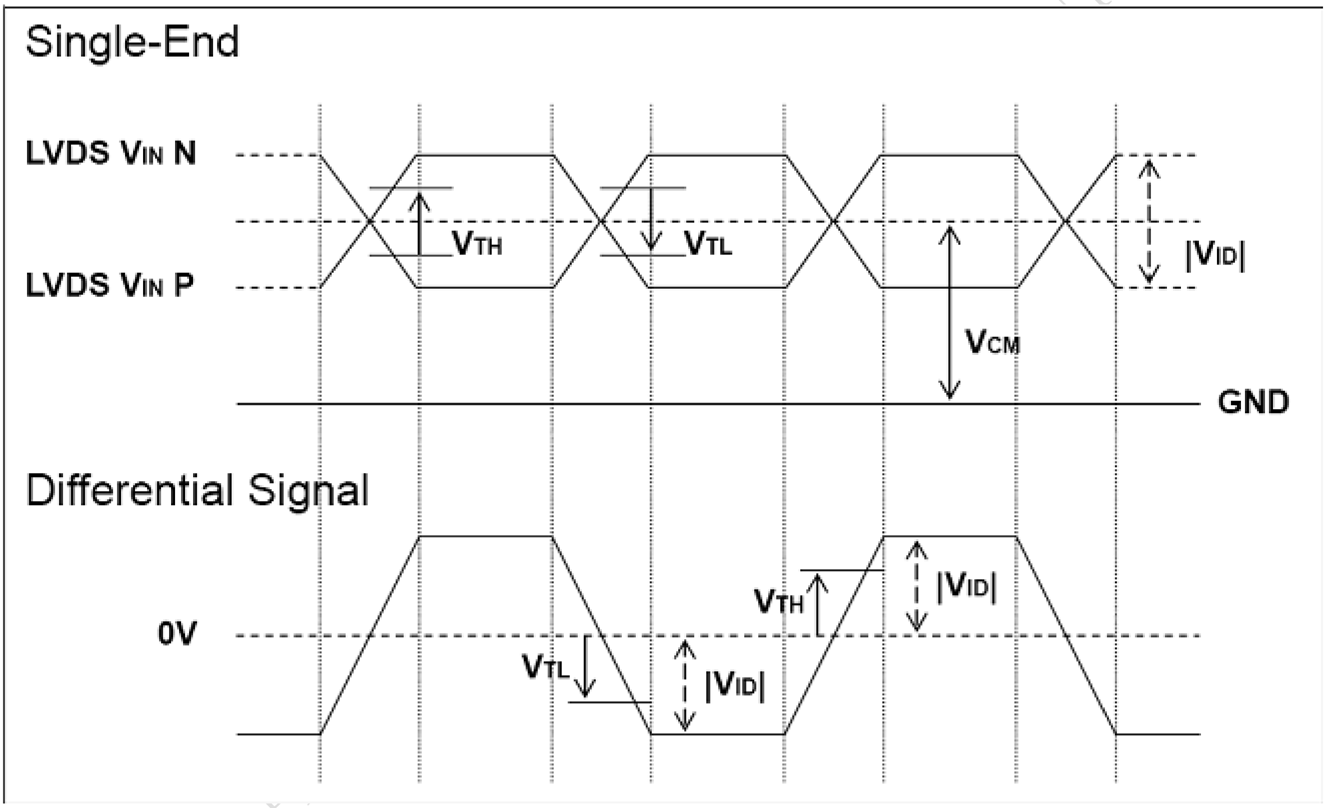


5.1.2 Signal Electrical Characteristics

Characteristics of each signal are as follows:

Symbol	Parameter	Min	Typ	Max	Units	Condition
VTH	Differential Input High Threshold	-		100	[mV]	VICM = 1.2V Note 1
VTL	Differential Input Low Threshold	-100		-	[mV]	VICM = 1.2V Note 1
∣VID∣	Input Differential Voltage	100	400	600	[mV]	Note 1
VICM	Differential Input Common Mode Voltage	1.125	-	1.375	[V]	VTH-VTL = 200MV (max) Note 1

Note 1: LVDS Signal Waveform



5.2 Backlight Unit

5.2.1 LED Driver

Following characteristics are measured under a stable condition at 25 °C (Room Temperature):

I_F	LED Forward Current		55		mA	Ta = 25 °C
V_F	LED Forward Voltage	-	2.85	3.3	Volt	I_F =55mA, Ta = 25°C
P_{LED}	LED Power Consumption	-	10.03	11.62	Watt	I_F =55mA, Ta = 25°C
LED Life Time		50,000			Hrs	I_F =55mA, Ta= 25°C

Note 1: Ta means ambient temperature of TFT-LCD module.

Note 3: I_F, V_F , P_{LED} are defined for single LED.

Note 4: If G215HAT02.1 module is driven by high current or at high ambient temperature & humidity condition. The LED life will be reduced.

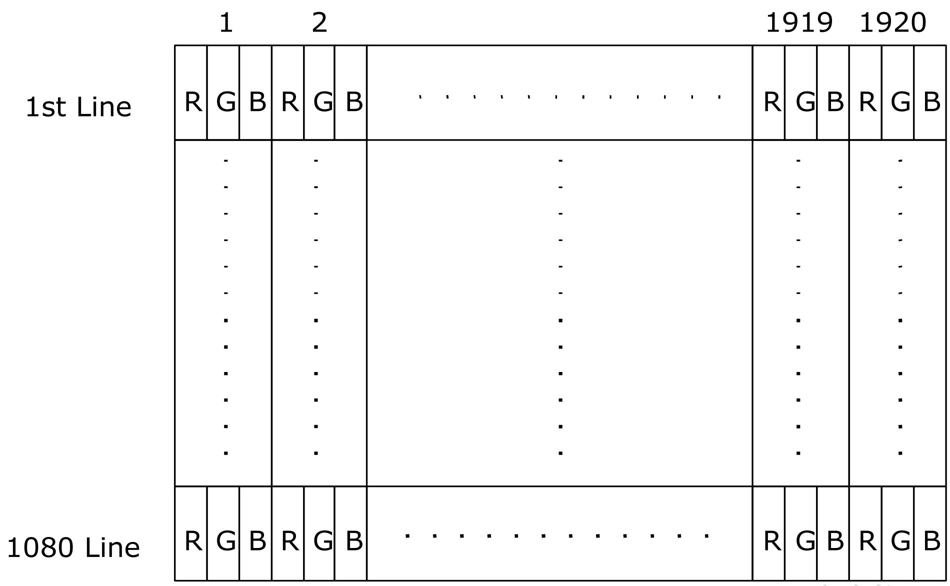
Note 5: Definition of life time: LED brightness becomes 50% of its original value. The minimum life time of LED unit is defined at the condition of I_F =55mA, Ta = 25°C (Room temperature)

Note 6: Each LED light bar consists of 64 pcs LED package (4 strings x 16 pcs / string)

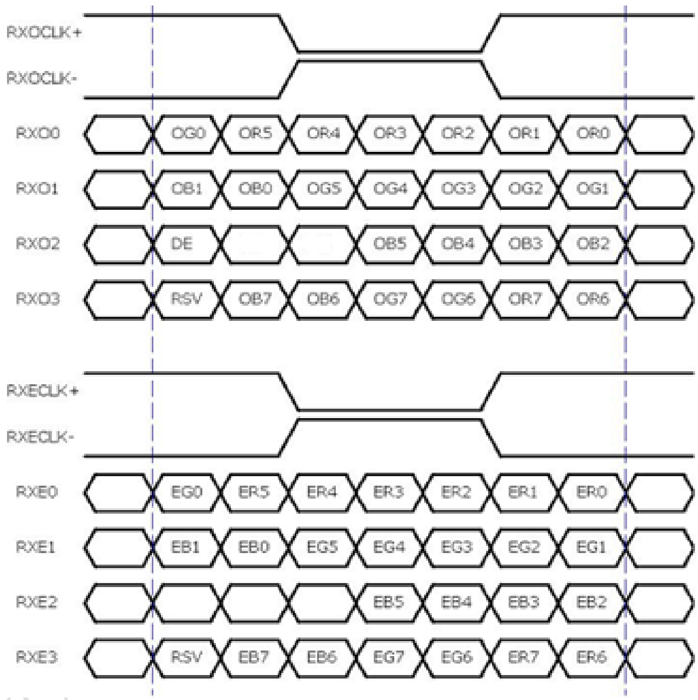
6. Signal Characteristic

6.1 Pixel Format Image

Following figure shows the relationship of the input signals and LCD pixel format.



6.2 The input data format



8 Bit Color Bit Order			
MSB	R7	G7	B7
	R6	G6	B6
	R5	G5	B5
	R4	G4	B4
	R3	G3	B3
	R2	G2	B2
	R1	G1	B1
LSB	R0	G0	B0

Note1: Please follow VESA.

Note2: 8-bits signal input.

6.3 Signal Description

The module uses a LVDS receiver embedded in AUO's ASIC. LVDS is a differential signal technology for LCD interface and a high-speed data transfer device.

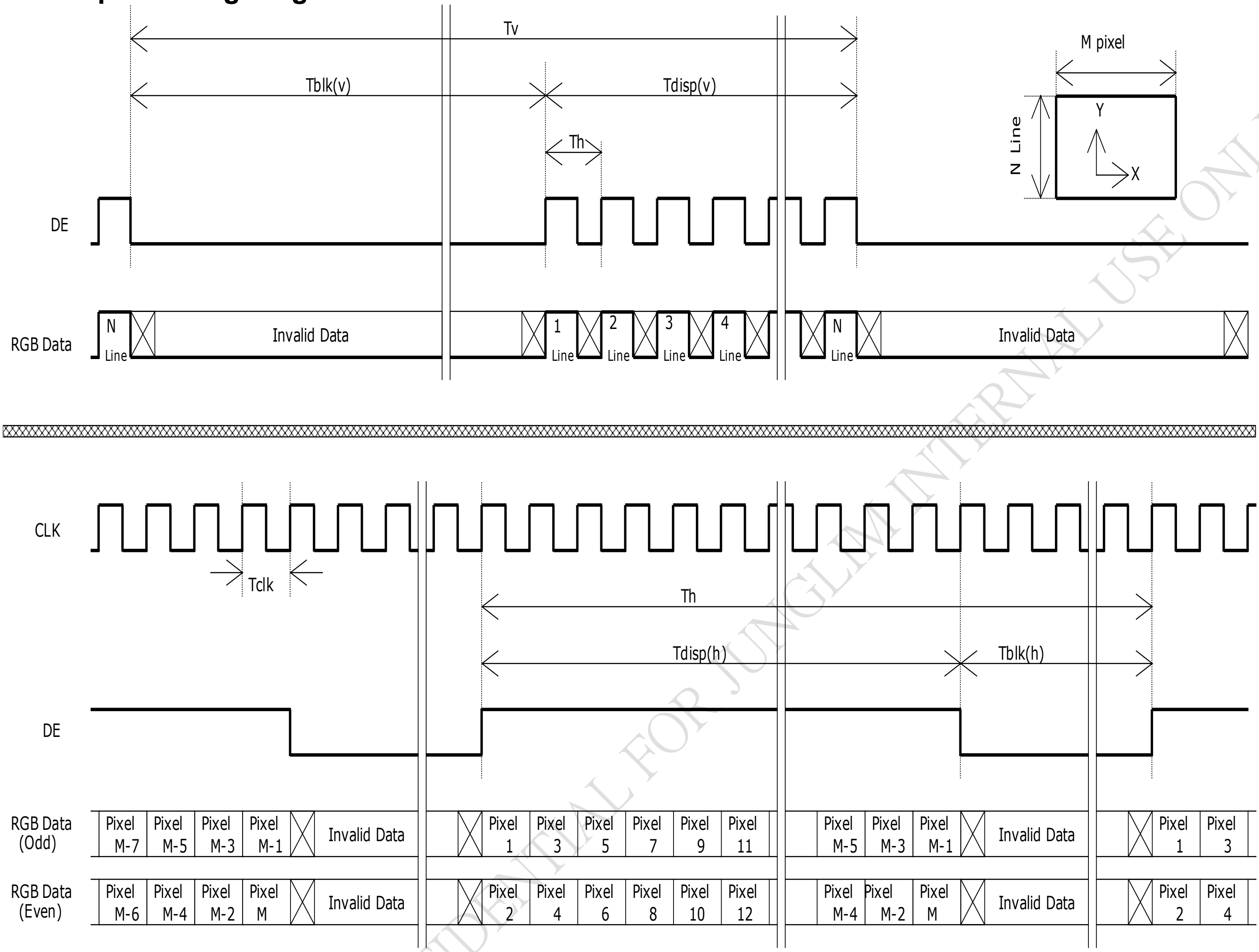
6.4 Interface Timing

6.4.1 Timing Characteristics

Signal	Item	Symbol	Min	Typ	Max	Unit
V-section	Period	Tv	1094	1130	1836	Th
	Active	Tdisp(v)	1080	1080	1080	Th
	Blanking	Tbp(v)+Tfp(v)+PWvs	14	50	756	Th
H-section	Period	Th	1000	1050	1678	Tclk
	Active	Tdisp(h)	960	960	960	Tclk
	Blanking	Tbp(h)+Tfp(h)+PWhs	40	90	718	Tclk
Clock	Period	Tclk	11.2	14.0	19.4	ns
	Frequency	Freq.	53.7	71.2	90.0	MHz
Frame Rate	Frame Rate	1/Tv	50	60	75	Hz

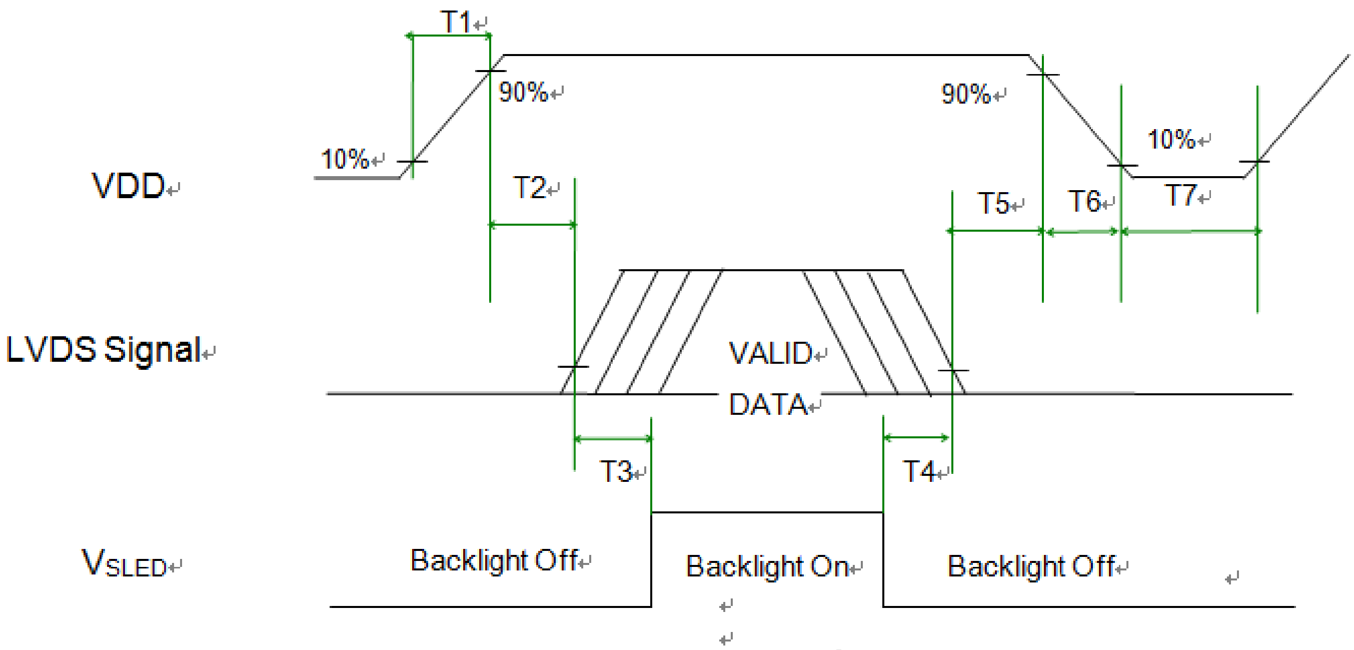
Note 1: The performance of the electro-optical characteristics may be influenced by variance of the vertical refresh rates.
Note 2: Horizontal period should be even.

6.4.2 Input Timing Diagram



6.5 Power ON/OFF Sequence

VDD power and LED on/off sequence are as follows. Interface signals are also shown in the chart. Signals from any system shall be Hi-Z state or low level when VDD is off.



Power ON/OFF sequence timing

Parameter	Value			Units
	Min.	Typ.	Max.	
T1	0.5	-	10	[ms]
T2	0	-	50	[ms]
T3	500	-	-	[ms]
T4	100	-	-	[ms]
T5	0		50	[ms]
T6	0	-	200	[ms]
T7	1000	-	-	[ms]

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7. Connector & Pin Assignment

Physical interface is described as for the connector on module. These connectors are capable of accommodating the following signals and will be following components.

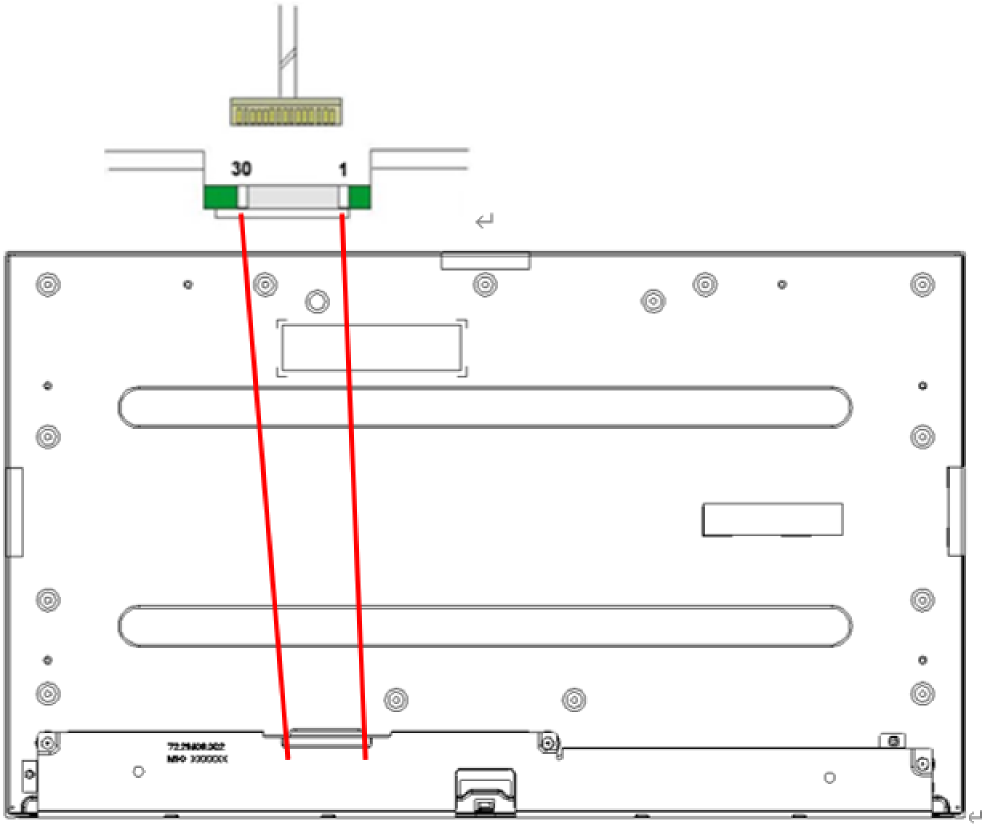
7.1 TFT LCD Module

TFT-LCD Connector	Manufacturer	SIN SHENG TERMINAL & MACHINE INC(STM)
	Part Number	MSBKT2407P30HB
Mating Connector	Manufacturer	JAE or Compatible
	Part Number	FI-X30HL (Locked Type) or Compatible

7.1.1 Pin Assignment

PIN #	Symbol	Description	Remark
1	RxO0-	Negative LVDS differential data input (Odd data)	
2	RxO0+	Positive LVDS differential data input (Odd data)	
3	RxO1-	Negative LVDS differential data input (Odd data)	
4	RxO1+	Positive LVDS differential data input (Odd data)	
5	RxO2-	Negative LVDS differential data input (Odd data)	
6	RxO2+	Positive LVDS differential data input (Odd data)	
7	GND	Ground	
8	RxOCLK-	Negative LVDS differential clock input (Odd clock)	
9	RxOCLK+	Positive LVDS differential clock input (Odd clock)	
10	RxO3-	Negative LVDS differential data input (Odd data)	
11	RxO3+	Positive LVDS differential data input (Odd data)	
12	RxE0-	Negative LVDS differential data input (Even data)	
13	RxE0+	Positive LVDS differential data input (Even data)	
14	GND	Ground	
15	RxE1-	Negative LVDS differential data input (Even data)	
16	RxE1+	Positive LVDS differential data input (Even data)	
17	GND	Ground	
18	RxE2-	Negative LVDS differential data input (Even data)	
19	RxE2+	Positive LVDS differential data input (Even data)	
20	RxECLK-	Negative LVDS differential clock input (Even clock)	
21	RxECLK+	Positive LVDS differential clock input (Even clock)	
22	RxE3-	Negative LVDS differential data input (Even data)	
23	RxE3+	Positive LVDS differential data input (Even data)	
24	NC	No connection (for internal use only. Do not connect)	
25	NC	No connection (for internal use only. Do not connect)	
26	NC	No connection (for internal use only. Do not connect)	

27	NC	No connection (for internal use only. Do not connect)	
28	VDD	Power Supply Input +5.0 Voltage	
29	VDD	Power Supply Input +5.0 Voltage	
30	VDD	Power Supply Input +5.0 Voltage	

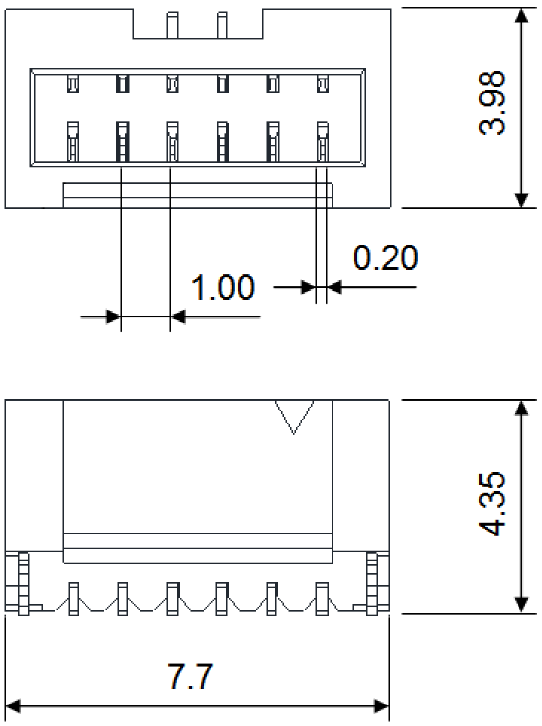


7.2 Backlight Unit: LED Connector

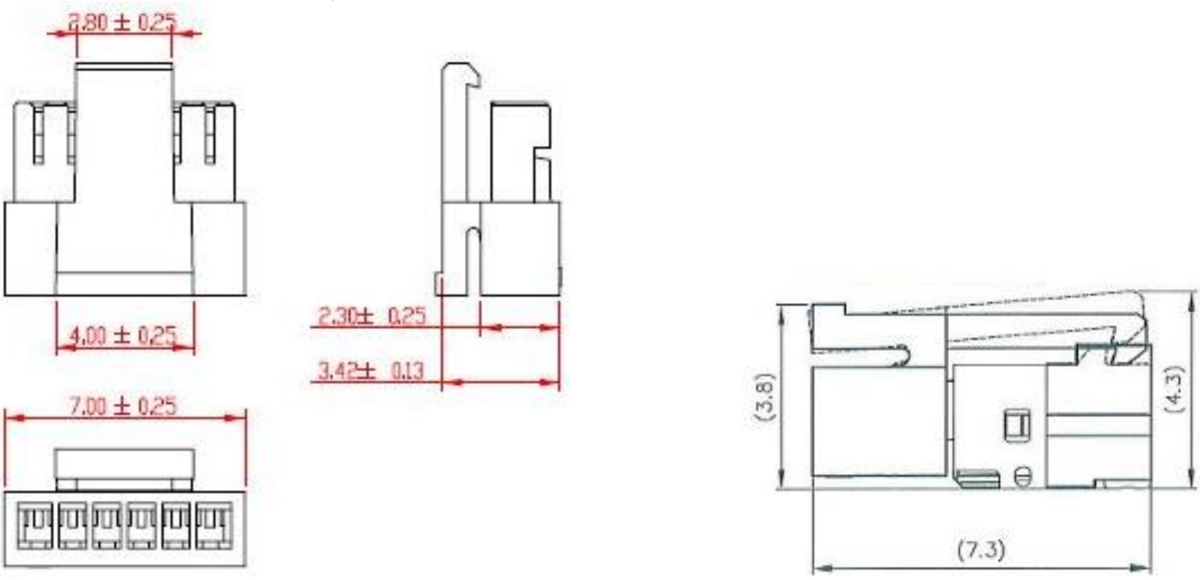
Backlight Connector	Manufacturer	CVILUX
	Part Number	CI1406M1VLD-NH
Mating Connector	Manufacturer	CVILUX
	Part Number	CI1406S0000-NH (Non-Locking type) CI1406SL000-NH (Locking type)

Backlight Connector dimension:

$H \times V \times D = 7.7 \times 3.98 \times 4.35, Pitch = 1.0(unit = mm)$

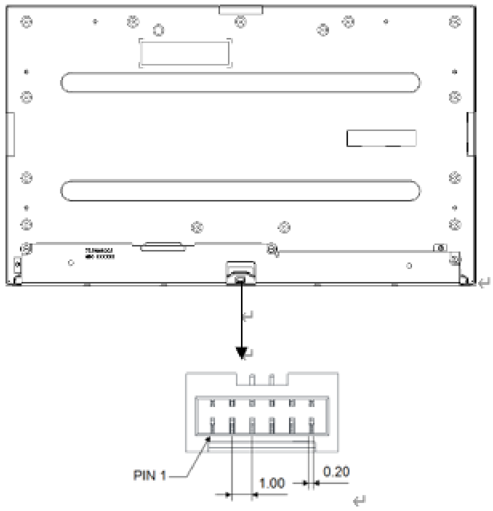


Mating Connector dimension:



7.3 LED Driver Connector Pin Assignment

Pin#	Symbol	Description	Remark
1	Ch1	LED Current Feedback Terminal (Channel 1)	
2	Ch2	LED Current Feedback Terminal (Channel 2)	
3	V_{SLED}	LED Power Supply Voltage Input Terminal	
4	V_{SLED}	LED Power Supply Voltage Input Terminal	
5	Ch3	LED Current Feedback Terminal (Channel 3)	
6	Ch4	LED Current Feedback Terminal (Channel 4)	



7.4 Touch Driver Connector

7.4.1 Touch Driver Connector

Name / Designation	TP controller
Manufacturer	EETI
Type / Part Number	EXC81H84

7.4.2 Connect & Pin Assignment

Manufacturer	ACSE or Compatible
Connector Model Number	wire to board / (50224-00501-001) or Compatible

USB Interface	
Pin No.	Designation
1	GND
2	VDD
3	GND
4	D+
5	D-

8. Reliability Test Criteria

Environment test conditions are listed as following table.

Items	Required Condition	Note
Temperature Humidity Bias (THB)	Ta= 50℃, 80%RH, 300hours	
High Temperature Operation (HTO)	Ta= 60℃, 300hours	
Low Temperature Operation (LTO)	Ta= 0℃, 300hours	
High Temperature Storage (HTS)	Ta= 60℃, 300hours	
Low Temperature Storage (LTS)	Ta= -20℃, 300hours	
Vibration Test (Non-operation)	Acceleration: 1.5 Grms Wave: Random Frequency: 10 - 200 Hz Duration: 30 Minutes each Axis (X, Y, Z)	
Shock Test (Non-operation)	Acceleration: 50 G Wave: Half-sine Active Time: 20 ms Direction: ±X, ±Y, ±Z (one time for each Axis)	
Drop Test	Height: 46 cm, package test	
Thermal Shock Test (TST)	-20℃/30min, 60℃/30min, 100 cycles	*1*
On/Off Test	On/10sec, Off/10sec, 30,000 cycles	
ESD (Electro Static Discharge)	Contact Discharge: ± 8KV, 150pF(330Ω) 1sec, 8 points, 25 times/ point.	*2*
	Air Discharge: ± 15KV, 150pF(330Ω) 1sec 8 points, 25 times/ point.	

Note 1: The TFT-LCD module will not sustain damage after being subjected to 100 cycles of rapid temperature change. A cycle of rapid temperature change consists of varying the temperature from -20oC to 60oC, and back again. Power is not applied during the test. After temperature cycling, the unit is placed in normal room ambient for at least 4 hours before power on.

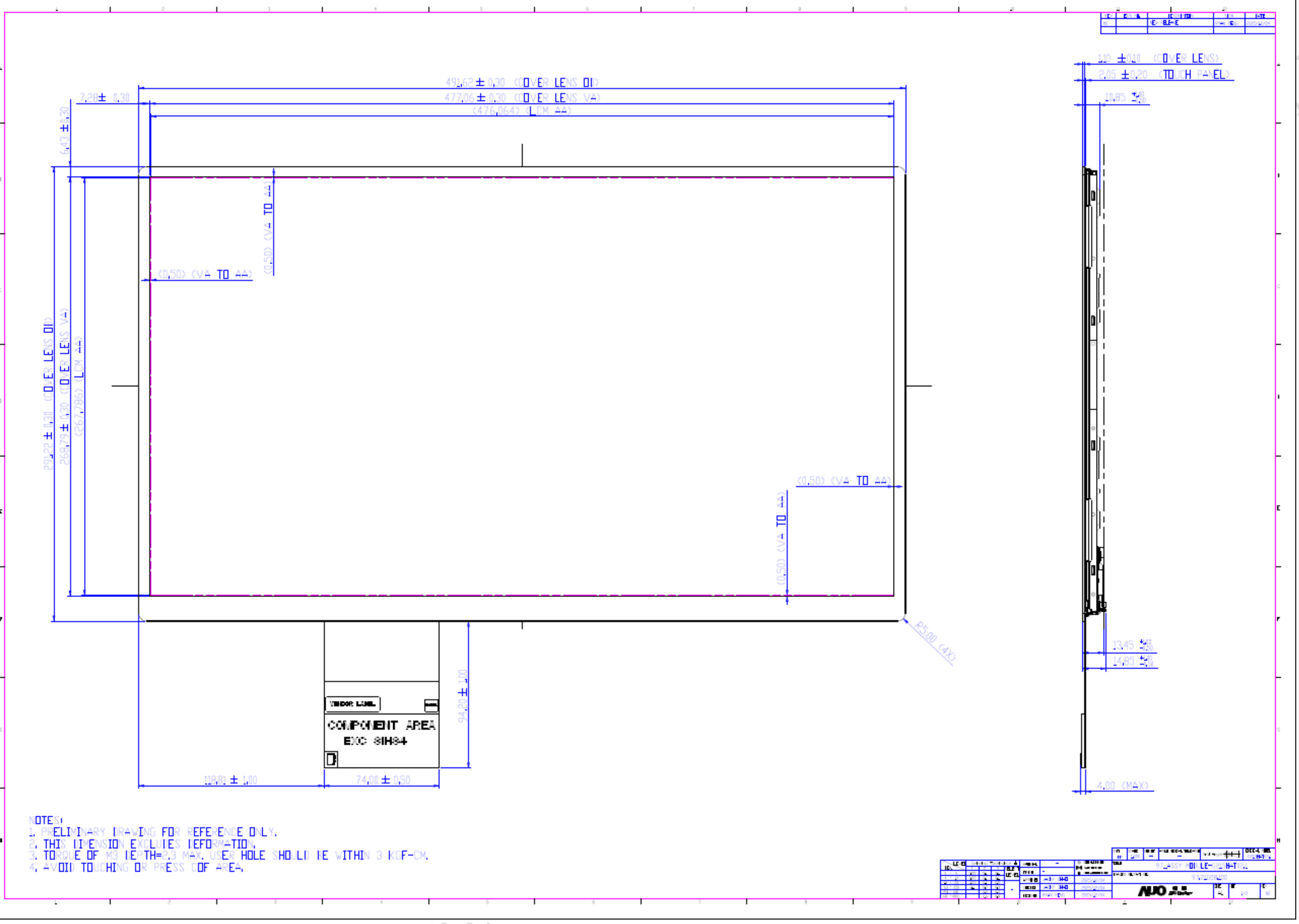
Note 2: According to EN61000-4-2 , ESD class B: Some performance degradation allowed. No data lost. Self-recoverable. No hardware failures.

Note 3: To inspect TFT-LCD module after reliability test, please store it at room temperature and room humidity for 24 hours at least in advance.

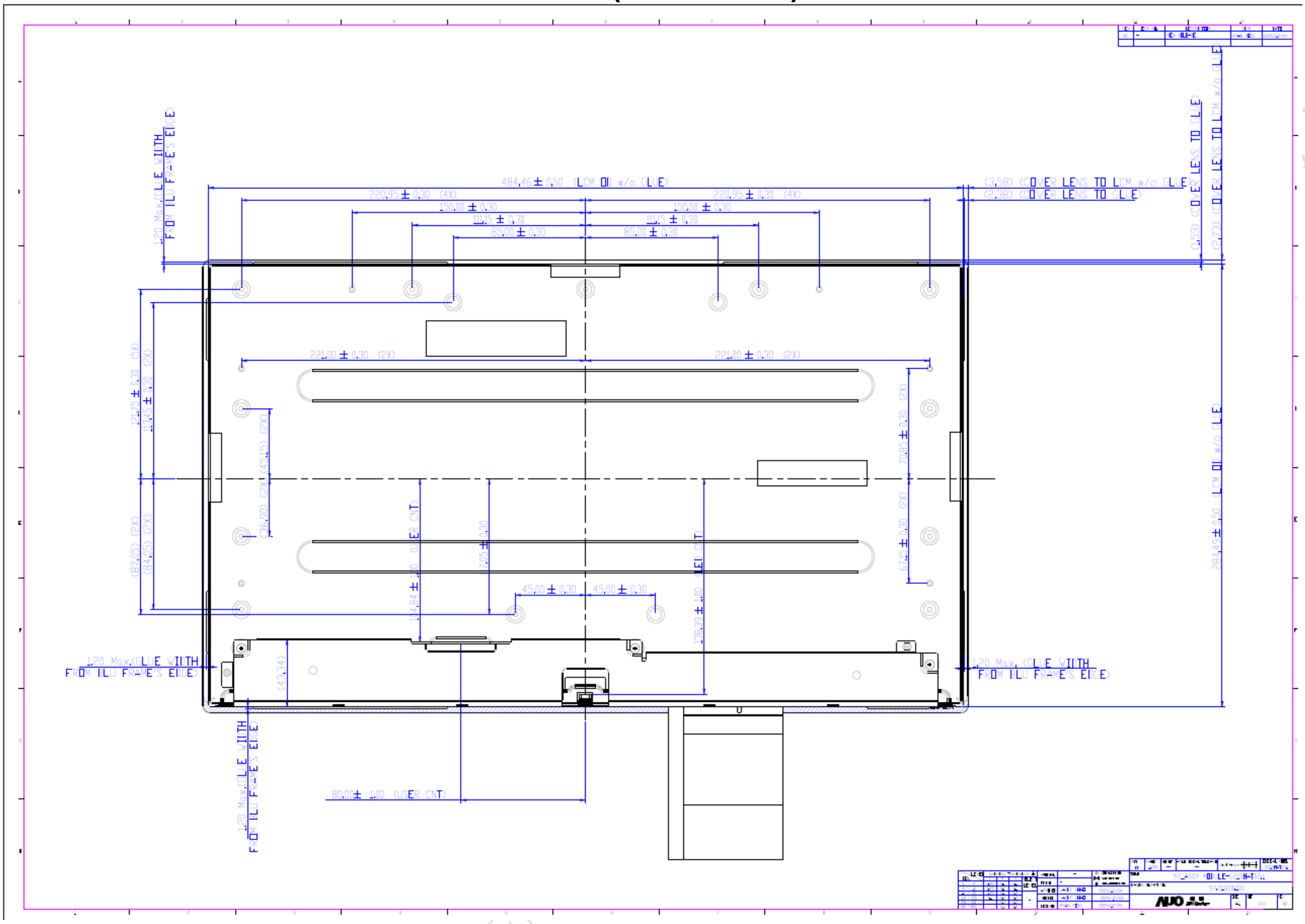
Note 4: The high temperature test item: THB & HTO & HTS & Thermal Shock, Mura shall be ignored.

9. Mechanical Characteristics

9.1 Total solution Outline Dimension (Front View)



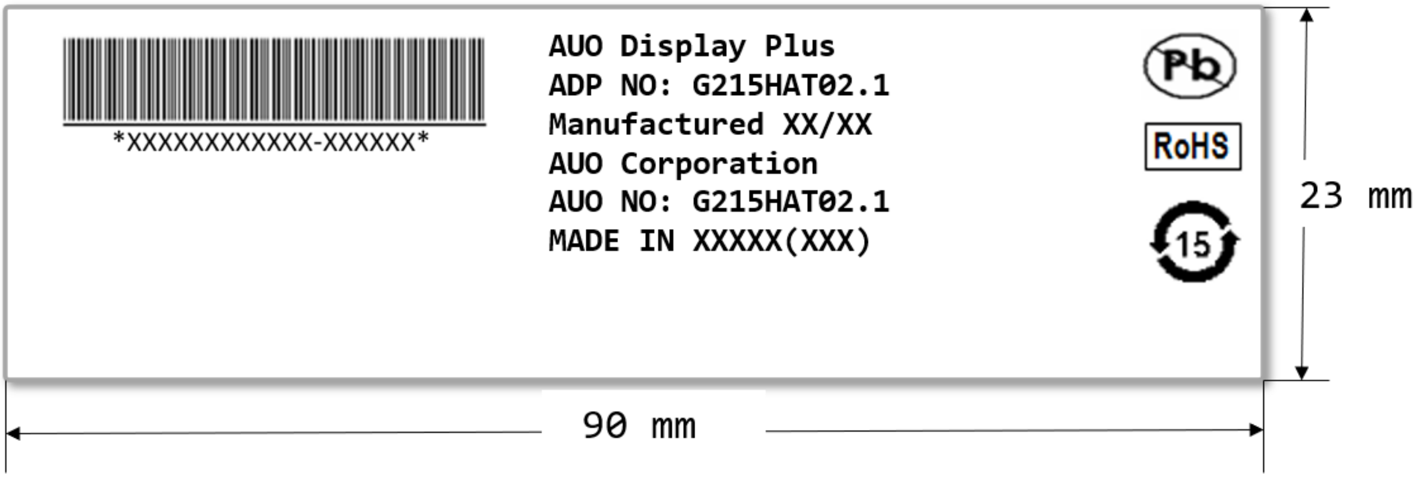
9.2 Total solution Outline Dimension (Back View)



10. Label and Packaging

10.1 Shipping Label

The label is on the panel as shown below:



Note 1: For Pb Free products, AUO will add for identification.

Note 2: For RoHS compatible products, AUO will add for identification.

Note 3: For China RoHS compatible products, AUO will add for identification.
Note 4: The Green Mark will be presented only when the green documents have been ready by AUO Internal Green Team.

10.2 Carton Package

TBD

11. Safety

11.1 Keen Edge Requirements

There will be no sharp edges or comers on the display assembly that could cause injury.

11.2 Materials

11.2.1 Toxicity

There will be no carcinogenic materials used anywhere in the display module. If toxic materials are used, they will be reviewed and approved by the responsible AUO toxicologist.

11.2.2 Flammability

The printed circuit board will be made from material rated 94-V1 or better. The actual UL flammability rating will be printed on the printed circuit board.

11.3 Capacitors

If any polarized capacitors are used in the display assembly, provisions will be made to keep them from being inserted backwards.

11.4 International Safety Standard Compliance

The TFT-LCD module will satisfy all requirements for compliance to:IEC/UL 62368-1